Student Reference Number:

DO NOT REMOVE FROM EXAM VENUE

University of Plymouth

MODULE CODE:	**ELEC345**
TITLE OF PAPER:	**High Speed Communications**

TIME ALLOWED	**3 hours**
DATE	**Tuesday 24th January 2023**
TIME	**09.00 AM – 12.00 PM**
FACULTY	**Science and Engineering**
SCHOOL	**Engineering, Computing and Mathematics**
ACADEMIC YEAR	**2022/23**
STAGE	**Three**

INSTRUCTIONS TO CANDIDATES:

Candidates should attempt **four out of the five** questions.
All questions carry equal marks. Marks for parts of questions are shown where appropriate.

Data provided:
Filter Tables
Equation list
Four full size Smith Charts – please attach completed exam paper to answer book.

Write student reference number on the top of each page and relevant question number on top of pages 10 to 13

Candidates are not permitted to look at the examination paper until instructed to do so.

Student Reference Number:

1. This question is on propagation and link budgets.

(a) You have been asked by your lecturer to setup a point-to-point Wi-Fi link to his house in Saltash. You have found a package that contains two high gain antennae, a receiver, and a variable output transmitter. You find the following: Distance = 8km. Transmit power = 0 to 27dBm. Antenna gains, G_T = 13dBi, G_R = 13dBi. Frequency = 5.8GHz. Minimum received power -75dBm. Maximum range of system = 15km.

(i) Calculate the path loss and received power

[5 marks]

(ii) Calculate the margin, how much the received power exceeds or falls short of the -75dB requirement for operation.

[5 marks]

(iii) Calculate the maximum range of the system, does it match the actual specification of 15km?

[5 marks]

(iv) What would be the benefits and disadvantages of switch to the 2.4GHz frequency for the link?

[5 marks]

(b) Given a channel bandwidth of 20Mhz and a noise figure of 1dB at the transmitter and receiver:

(i) Calculate the noise power in the channel and calculate the signal to noise ratio at the receiver.

[5 marks]

2. This question is on transmission lines.

(a) Sketch the equivalent circuit model for a transmission line and explain why the velocity of propagation is less than that of free space?

[5 marks]

(b) Why are transmission lines needed and at what frequency do they become necessary? (What is the problem with using a normal wire)

[5 marks]

(c) Given a transmission line with a characteristic impedance, Z_0 of 50Ω and source voltage of Vs = 12V, source impedance of Zs = 30Ω and Z_L = 150Ω. Calculate:

(i) The initial voltage on the line.

(ii) The voltage the line will settle at.

(iii) These voltages if both the load and source were matched

[6 marks]

(d) Using the same values as in part c. calculate the voltage at the load, Z_L when three reflections have occurred, Z_L, Zs and Z_L and sketch a bounce diagram showing all voltages.

[9 marks]

3. This question is about smith charts and matching networks.

(a) Plot the load Z_L = 20 + 25j ohms and:

(i) Find the voltage reflection coefficient

(ii) The voltage standing wave ratio (SWR) linear and in dB

(iii) The return loss and actual power that will be transmitted/radiated.

(iv) Plot all values of equal SWR.

[8 marks]

Question 3 continued …

Question 3 continued

(b) Plot the load $Y_L = 0.02 - 0.04j$ siemens and change it to an impedance on the chart and read it off.

[2 marks]

(c) Design both possible L-matches for a $Z_L = 15 - 70j$ ohm antenna sketching your circuit and labelling all values.

[10 marks]

(d) Choose one of your L matches and calculate the component values for a frequency of 200MHz.

[5 marks]

4. This question is about stub matching.

(a) Design a series single stub match for a load with an impedance of $Z_L = 100 + 50j$ ohms. Include your smith chart with your answers and show all workings and calculations.

[10 marks]

(b) Design a parallel double stub match 0.04λ from an impedance of $Z_L = 120 - 130j$ ohms. The spacing between the two stubs is set at 3/8λ. Include your smith chart with your answers and show all workings and calculations.

[15 marks]

5. This question is about filters.

(a) Explain how a filter, especially a digital filter can be used to model a multipath channel?

[4 marks]

Question 5 continued …

Question 5 continued

(b) What is spatial filtering and where is it used/found?

[4 marks]

(c) Using the tables provided design a passive ladder network to realise a Butterworth filter with F_c = 2 MHz and F_s = 6 MHz to achieve A_{min} = 45 dB between a 50Ω source and load.

[7 marks]

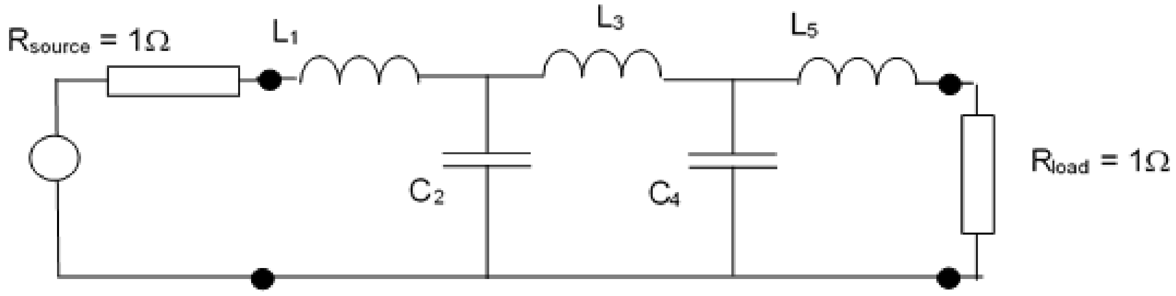
(d) Using the tables provided design an active high pass Butterworth filter with F_s = 2 MHz and F_c = 8 MHz and A_{min} = 60 dB and impedances of 10kΩ

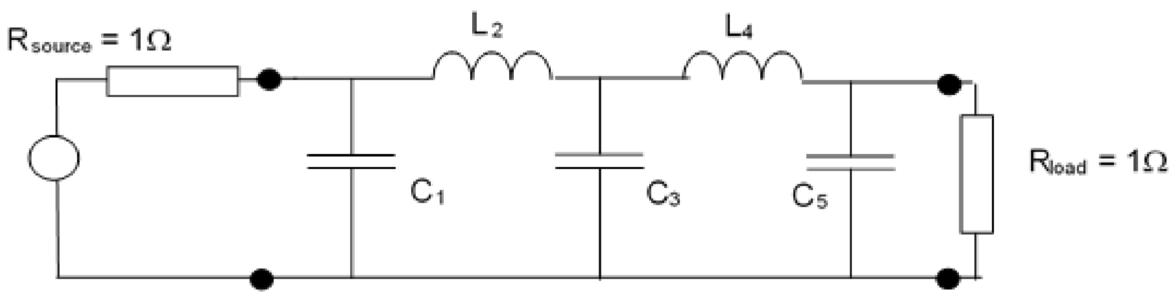
[10 marks]

FILTER DESIGN TABLES

Normalised Butterworth ladder network element values

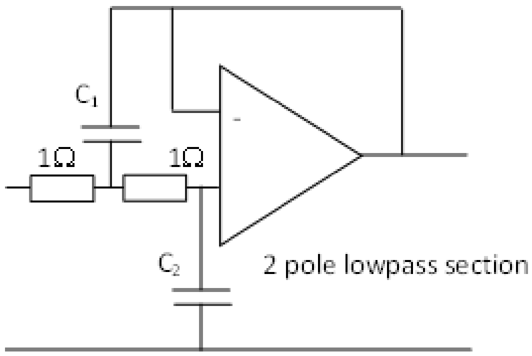
n	C_1	L_2	C_3	L_4	C_5	L_6	C_7
2	1.4142	1.4142					
3	1	2	1				
4	0.7654	1.8478	1.8478	0.7654			
5	0.618	1.618	2	1.618	0.618		
6	0.5176	1.4142	1.9319	1.9319	1.4142	0.5176	
7	0.445	1.247	1.8019	2	1.8019	1.247	0.445
	L_1	C_2	L_3	C_4	L_5	C_6	L_7

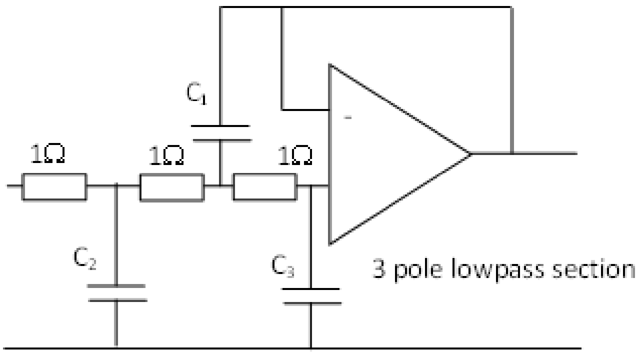




Normalised Butterworth active element values

n	C_1	C_2	C_3
2	1.414	0.7071	
3	3.546	1.392	0.2024
4	1.082 2.613	0.9241 0.309	
5	1.753 3.253	1.354 0.309	0.4214
6	1.035 1.414 3.863	0.966 0.7071 0.2588	
7	1.531 1.604 4.493	1.336 0.6235 0.2225	0.4885





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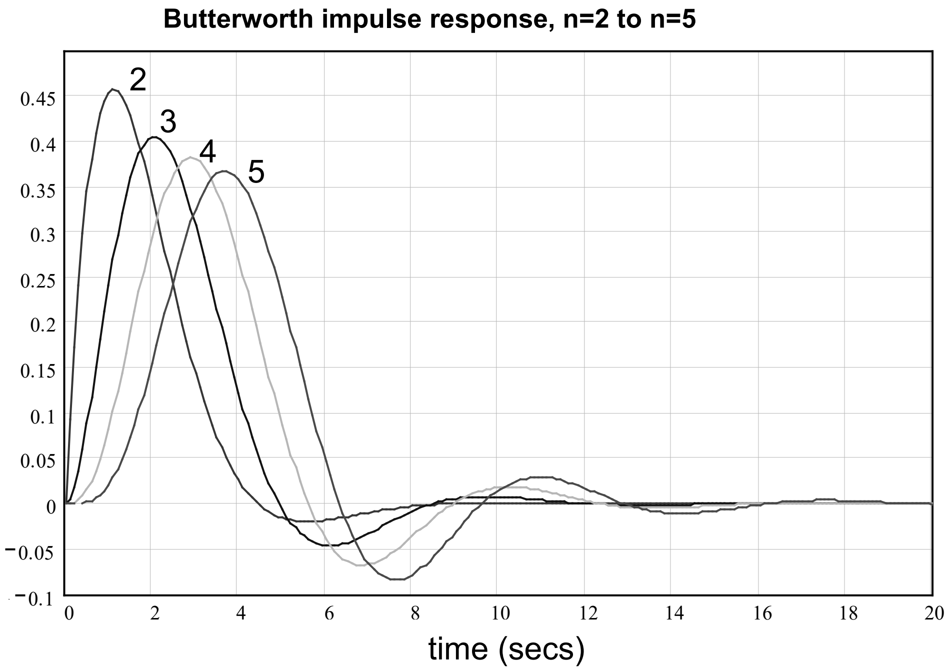
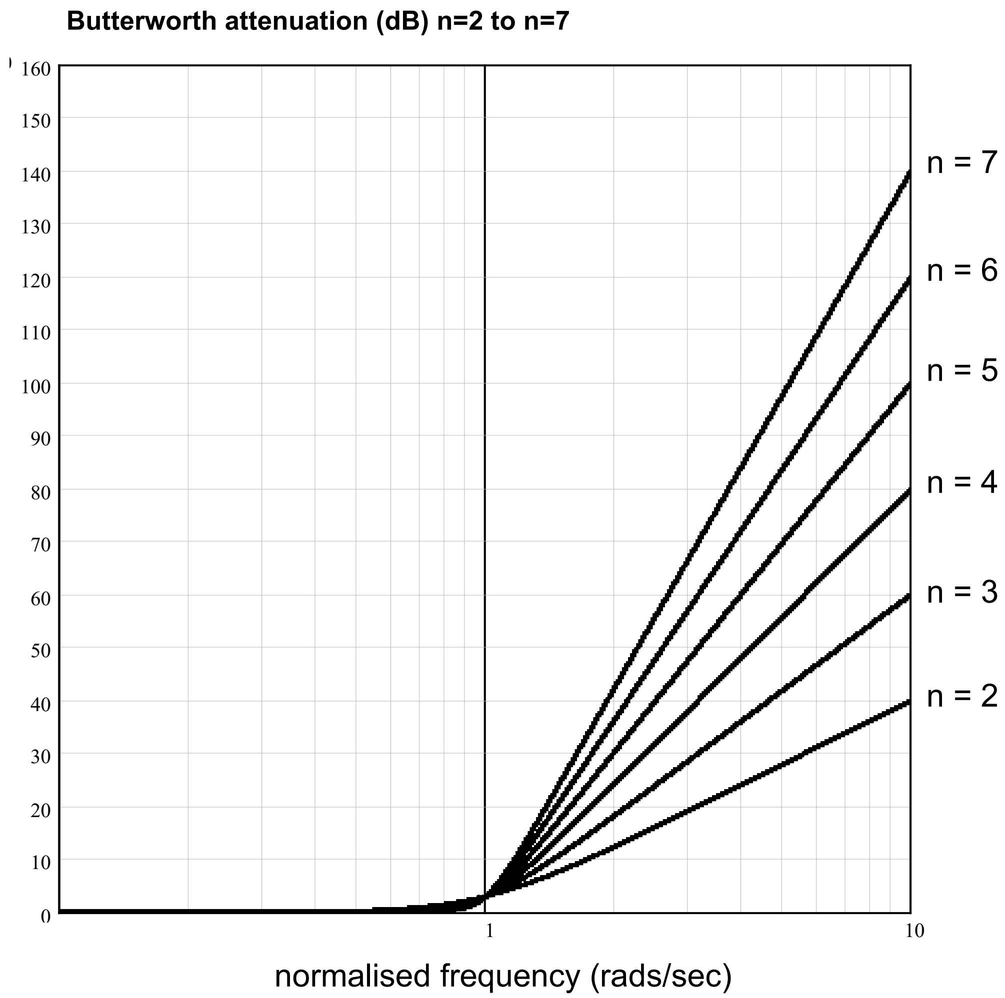
Normalised Butterworth impulse response and attenuation characteristics





Useful Equations

$$L \Rightarrow \frac{L.R_o}{2\pi F_c} \qquad C \Rightarrow \frac{C}{2\pi F_c R_o} \qquad R \Rightarrow R.R_o \; A_s = \frac{F_s}{F_c}$$

$$R_{HP} = \frac{1}{C_{LP}} \qquad and \qquad C_{HP} = \frac{1}{R_{LP}}$$

$$L = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right)\; dB$$

$$C = P_T + G_T - PEPL + G_R - \beta \quad \text{[dBm]}$$

$$PEPL = 20\log_{10}\left(\frac{R^2}{h_T h_R}\right) \quad \text{[dB] (plane earth path loss)}$$

$$\beta = 20 + \frac{f_{MHz}}{40} + 0.18L - 0.34H + K \quad \text{[dB]}$$

$$K = \begin{cases} 0.094U - 5.9 & \text{, for inner city areas} \\ 0 & \text{, elsewhere} \end{cases}$$

$$\left(f_d = \frac{v}{\lambda} = \frac{v}{c} f_{Hz} \right)$$

$$\gamma_{div,2} = \frac{\gamma_1 + \gamma_2 + 2\sqrt{\gamma_1 \gamma_2}}{2} \qquad \gamma_{div,N} = \sum_{i=1}^{N} \gamma_i$$

$$N = -174 + F + 10\log_{10}(B) \text{ dBm}$$

$$N = -204 + F + 10\log_{10}(B) \quad (dBW)$$

$\Gamma = \frac{Z - Z_0}{Z + Z_0}$ return loss = $10\log P_r/P_i = -20\log|\Gamma|$

VSWR = $1+|\Gamma|/1-|\Gamma|$

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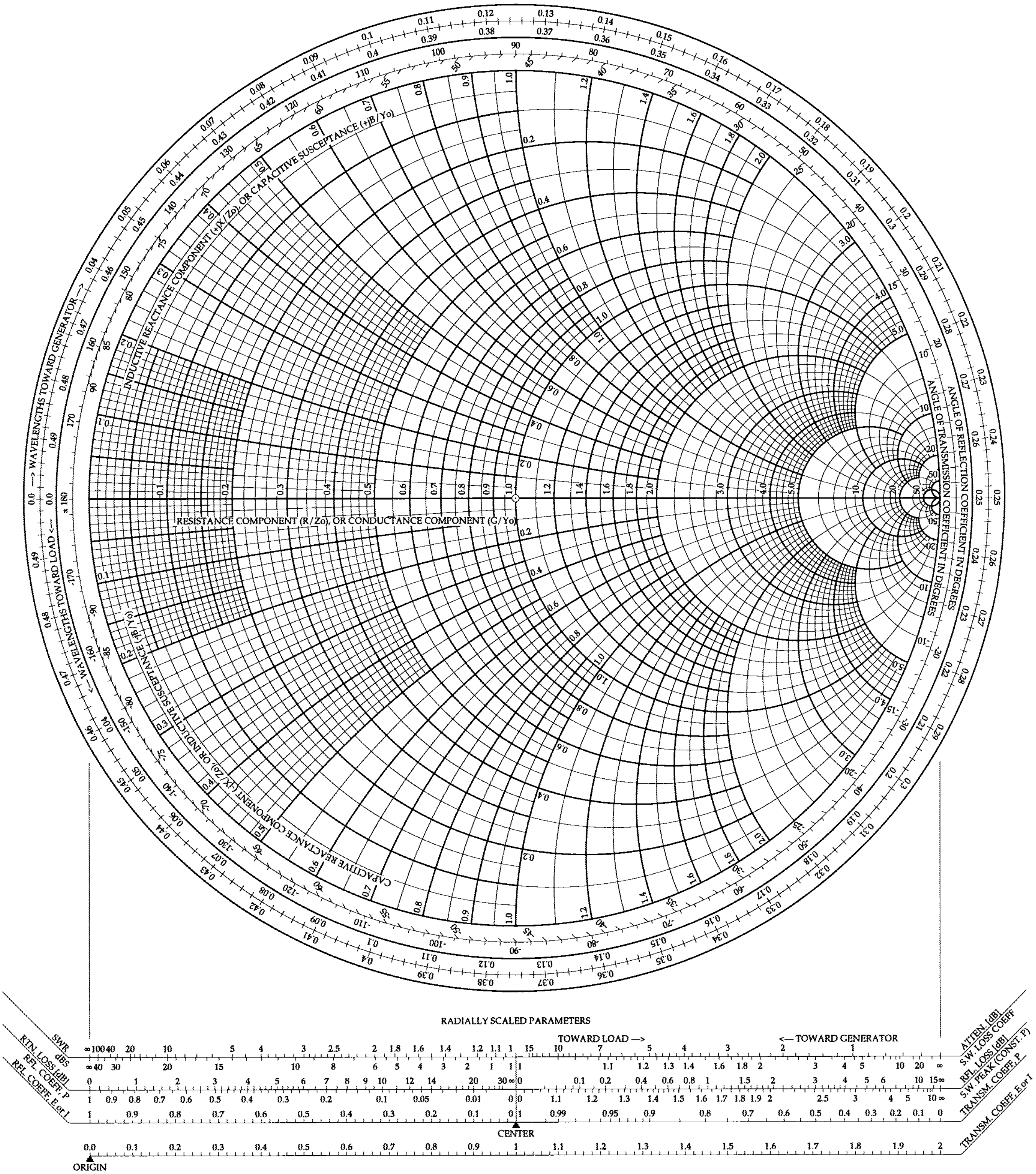
The Complete Smith Chart



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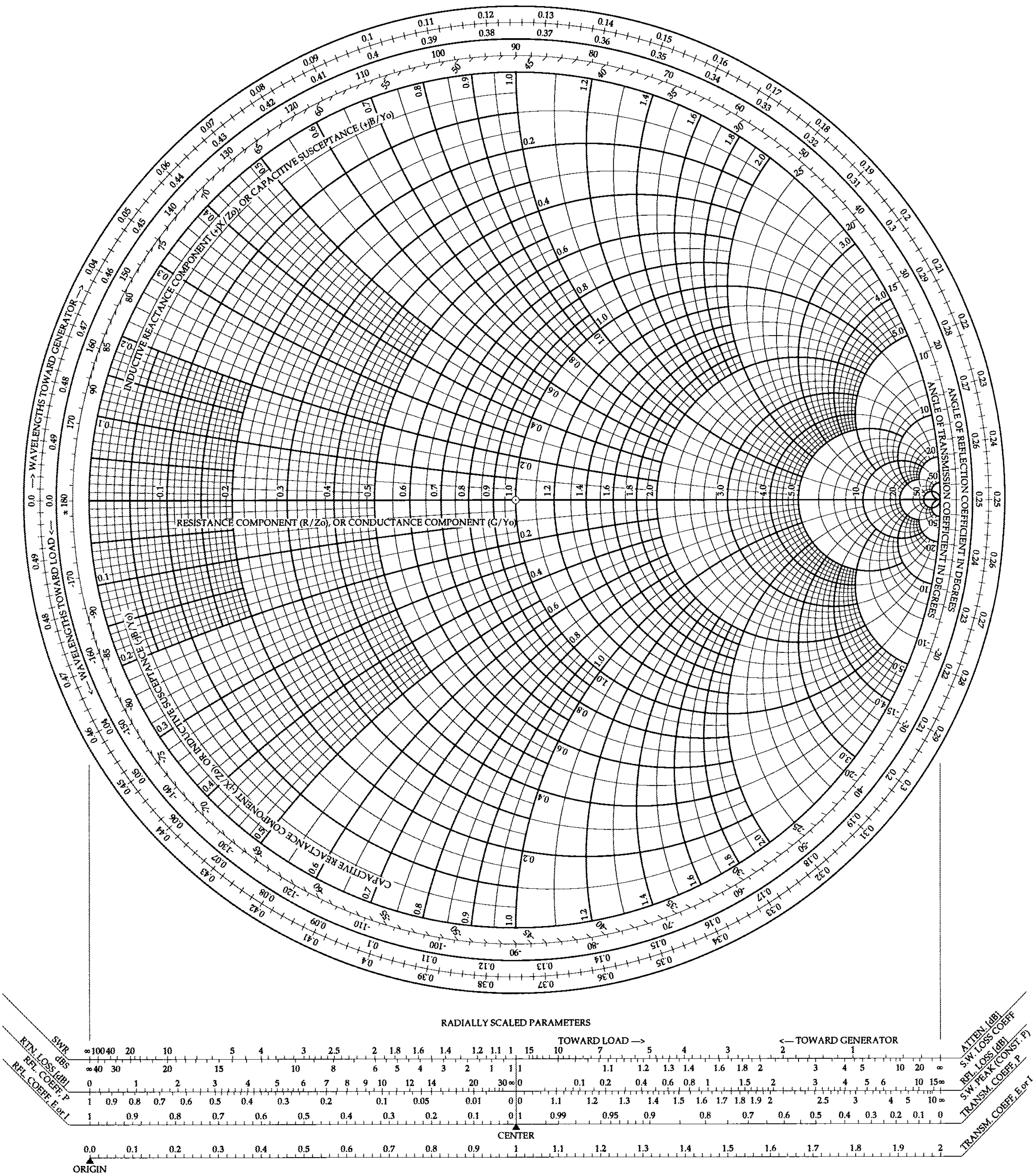

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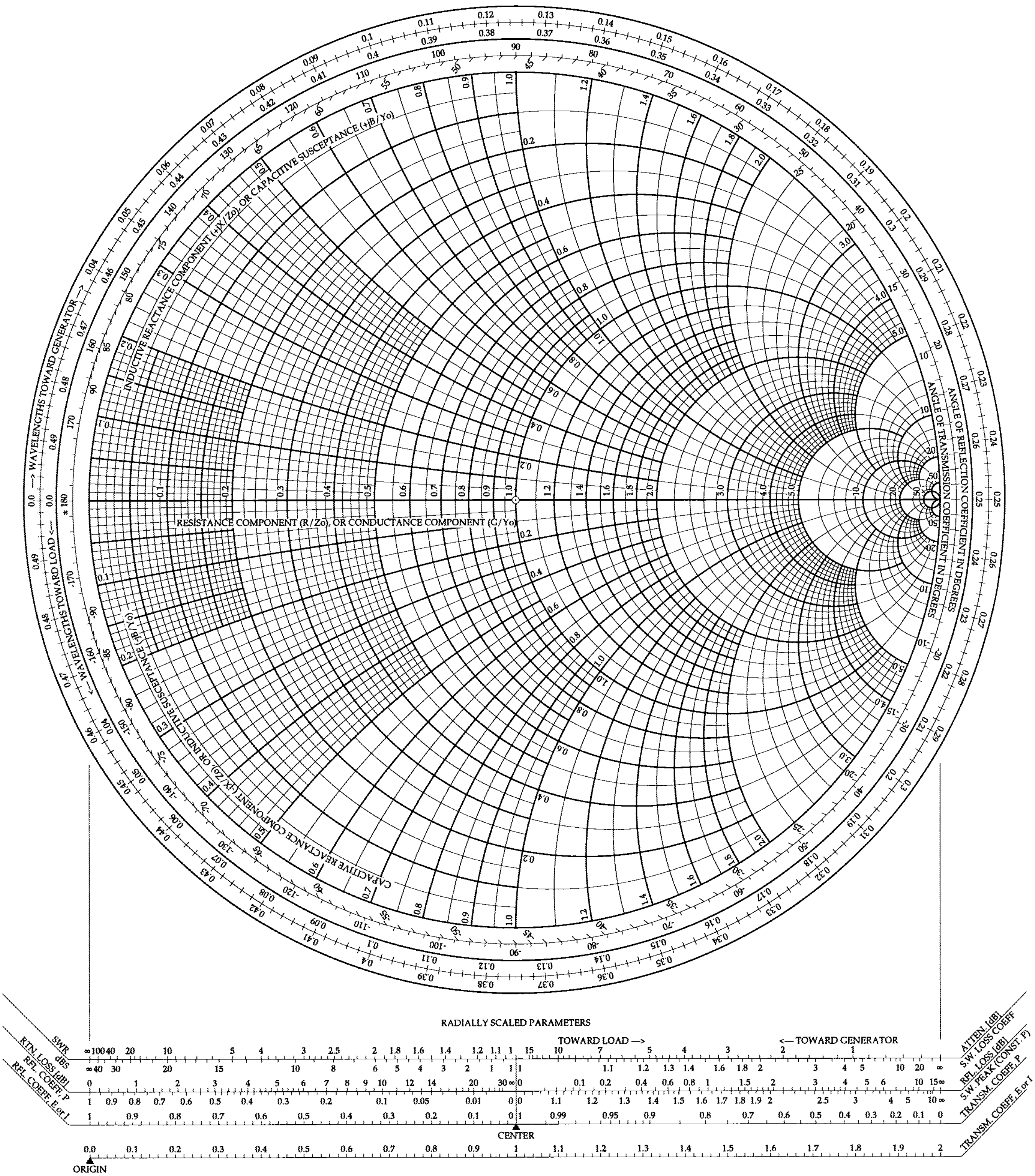

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The Complete Smith Chart